

Adaptive Context Aggregation Network With Prediction-Aware Decoding for Multimodal Brain Tumor Segmentation

Guanghui Yue^{ID}, *Member, IEEE*, Shangjie Wu^{ID}, Jie Du^{ID}, *Member, IEEE*,
Tianwei Zhou^{ID}, *Associate Member, IEEE*, Bin Jiang^{ID}, *Member, IEEE*,
and Tianfu Wang^{ID}

Abstract—Precise segmentation of brain tumor from magnetic resonance images (MRIs) is of vital importance for lesion measurement, surgical planning decision, and clinical treatment evaluation. In the context of rapid development of deep neural networks (DNNs), an increasing number of DNN-based brain tumor segmentation methods have been recently reported. However, most methods usually ignore the interaction among multimodal data and multiscale context features, resulting in poor segmentation accuracy on challenging cases, e.g., tumors with complex interactions between subregions. This article proposes a novel adaptive context aggregation network (ACANet) for multimodal brain tumor segmentation. ACANet follows a parallel encoder–decoder framework. Specifically, to bridge the gaps between different modalities, we propose an adaptive context aggregation (ACA) module that adaptively fuses multimodal features from the encoders of two input branches to explore their complementary information and aggregates multiscale context information extracted by a set of dilated convolutions. To cope with the complex boundary interactions, we propose a prediction-aware decoding strategy (PDS) at the decoder to help the network focus more on error-prone regions through the guidance of two temporal predictions when incorporating the features from the encoder. Extensive experiments on two brain tumor segmentation datasets demonstrate the superiority of ACANet over eight mainstream methods. Additionally, a series

of ablation studies supports the effectiveness of ACANet’s underlying concepts.

Index Terms—Brain tumor segmentation, deep neural network (DNN), multimodal data fusion, multiscale, prediction-aware decoding.

I. INTRODUCTION

BRAIN tumor, as a neurological tumor with highly aggressive and malignant characteristics, poses a profound threat to the physical and psychological health of patients [1]. It is often found in the later stages due to its high complexity, making the clinical treatment very difficult. Therefore, early diagnosis of brain tumor is critical. Clinically, magnetic resonance images (MRIs) are commonly used as a primary selection of monitoring and diagnosing brain tumors. MRIs include four imaging modalities, namely fluid-attenuated inversion recovery (FLAIR), T2-weighted (T2), T1-weighted (T1), and T1-with-contrast-enhanced (T1CE) [2], [3], [4]. These imaging modalities provide different information about brain tissue structure and lesions. By performing brain tumor segmentation on MRIs, it can help radiologists more accurately identify and localize tumor tissues, allowing for a better plan surgery and monitor treatment [5]. However, the task of brain tumor segmentation on MRIs by a radiologist is laborious and subjective, usually resulting in annotation uncertainty. Recently, automatic brain tumor segmentation on MRIs has attracted increasing attention in the context of rapid development of deep learning, showing great potentials in reducing the burden of radiologists and improving the accuracy of segmentation [6], [7].

Deep neural networks (DNNs) have obtained remarkable success recently in general-purpose medical image segmentation [8], [9], [10] and specific-purpose medical image segmentation [11], [12], [13], [14], [15], [16], [17], [18], [19] due to their powerful lesion-aware feature representation capabilities. For example, in the early research, Ronneberger et al. [20] build a U-shaped structure, named U-Net, and stack features layer-by-layer from the encoder to the decoder in a skip connection to compensate for the information loss induced by downsampling operations. Later, several variants of U-Net, e.g., Attention U-Net [21],

Manuscript received 28 December 2023; revised 29 April 2024; accepted 13 May 2024. Date of publication 2 July 2024; date of current version 8 July 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62371305; in part by the Natural Science Foundation of Shenzhen under Grant JCYJ20230808105906013; in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2024A1515030025, Grant 2024A1515030278, and Grant 2022A1515010160; in part by the Medicine Plus Program of Shenzhen University under Grant 2024YG007; and in part by the Tencent “Rhinoseros Birds”—Scientific Research Foundation for Young Teachers of Shenzhen University. The Associate Editor coordinating the review process was Dr. Hui Zhang. (Corresponding author: Jie Du.)

Guanghui Yue, Shangjie Wu, Jie Du, and Tianfu Wang are with the National-Regional Key Technology Engineering Laboratory for Medical Ultrasound, Guangdong Key Laboratory for Biomedical Measurements and Ultrasound Imaging, School of Biomedical Engineering, Shenzhen University Medical School, Shenzhen 518054, China, and also with the Marshall Laboratory of Biomedical Engineering, Shenzhen University, Shenzhen 518060, China (e-mail: yueguanghui@szu.edu.cn; 2310247015@email.szu.edu.cn; dujie@szu.edu.cn; tfwang@szu.edu.cn).

Tianwei Zhou is with the College of Management, Shenzhen University, Shenzhen 518060, China (e-mail: tianwei@szu.edu.cn).

Bin Jiang is with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao 266580, China (e-mail: jiangbin@upc.edu.cn).

Digital Object Identifier 10.1109/TIM.2024.3417601

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UNet++ [22], and UNet3+ [23], have been reported by updating U-Net from different aspects. Motivated by the success of transformer [24] in natural language processing, transformers have been introduced into the segmentation task to overcome the limitations of CNNs in capturing global information [25], [26]. In comparison to CNNs, transformers leverage multihead self-attention mechanisms to more effectively capture global information. For instance, Cao et al. [27] designed a U-shape framework based on transformer block, named Swin-Unet, for medical image segmentation, leveraging the global information learning capabilities of transformers to enhance segmentation performance. Gao et al. [28] updated the encoder and decoder of U-Net by transformer layers. To further improve the segmentation, they utilized a mutual supervision of the shallowest feature map and the deepest feature map. Hatamizadeh et al. [29] utilized vision transformer to augment feature representation for volumetric medical image segmentation. By combining the strengths of transformer and U-Net, attempts [30], [31] were also made in feature fusion and representation to improve the segmentation accuracy.

Compared to traditional medical image segmentation tasks, brain tumor segmentation on MRIs is more challenging and complex because more imaging modalities, e.g., FLAIR, T2, T1, and T1CE, should be comprehensively analyzed [32], [33], [34], [35], [36]. These image modalities not only hold their own unique information but also include complementary information for the same scene [37]. Effectively leveraging these modalities would contribute to good segmentation performance. However, how to effectively integrate these modalities is a key consideration in the design of brain tumor segmentation algorithms.

Recently, considering that different modalities represent the same scene from different perspectives, many methods have been reported for multimodal brain tumor segmentation [38], [39], [40], [41]. According to the fusion level of multimodal information, these methods can be categorized into three types: early-fusion, late-fusion, and multilevel fusion based methods. The former usually employs a simple cascaded operation on multimodal data before feeding them into the segmentation network [42], [43]. Late-fusion-based methods extract features for each modality separately using encoders and integrate the extracted high-level features from different modalities. The integrated features are subsequently fed into the decoder [44], [45]. The above methods only consider the fusion of multimodal data in the image level or later stage of the encoder, ignoring the interactions of multilevel features of the encoder.

Generally, different level features provide distinguished concepts for tumor representation, and effectively fusing multilevel features is conducive to accurate segmentation. In view of this, multilevel fusion-based methods have recently attracted increasing attention. For instance, Xing et al. [46] applied a multibranch encoder to extract multimodal features at multilevel. Subsequently, the top-level features are fed into a specifically designed module for feature fusion, while features of the other levels are cascaded along the channel direction and fed into a modality-sensitive

gating module for filtering modality-aware low-resolution features. Yang et al. [47] designed a transformer-based network to extract multimodal features and integrated these features by a cross-feature fusion module at different levels. Zhang et al. [45] proposed a mutual learning-based multimodal segmentation method that learns collaboratively to distill the commonality between high-level representations of different modalities. Wang et al. [48] proposed a multi-scale context-aware network for multimodal medical image segmentation by extracting context information with dense skip connection. More recently, works based on pixel-level fusion have also attracted researchers' attention [49], [50]. For example, Liu et al. [49] introduced a multitask model for accurate multimodal brain tumor segmentation, which takes pixel-level image fusion as the auxiliary task. Given that the edge information is very important for the segmentation task, Zhu et al. [51] proposed a novel brain segmentation method by jointly mining edge information and deep semantics in multimodal MRI. Although validated effectively in certain scenarios, these methods usually fail to precisely segment brain tumors in challenging cases due to the following limitations.

- 1) Most methods usually ignore the gaps between different modalities when fusing features at each level, which may mask discriminative information from different modalities.
- 2) Most methods produce segmentation uncertainty in boundary regions because of the low contrast between the tumor and nontumor regions in MRIs.

In this article, to advance this topic, we introduce a novel network, named adaptive context aggregation network (ACANet), for accurate multimodal brain tumor segmentation using multimodal 2-D MRIs slices. Our ACANet follows an encoder-decoder framework with two parallel network branches. Specifically, we categorize the MRIs into two groups (i.e., T1 and T1CE, and T2 and FLAIR) in view of imaging principles and feed them into two parallel branches, respectively. To bridge the gaps between different modalities, we propose an adaptive context aggregation (ACA) module that adaptively fuses multimodal features from the encoders of two feature extraction networks to explore their complementary information and aggregates cross-scale context information extracted by a set of dilated convolutions. To make the ACA module concentrate on tumor-aware regions, we combine the temporal predictions from two branches as the guidance information. To cope with the complex boundary interactions, we propose a prediction-aware decoding strategy (PDS) at the decoder to assist the network in concentrating on error-prone regions through the guidance of the difference between two temporal predictions when incorporating the features from the ACA modules. Experimental results on the BraTS 2020 and BraTS 2021 datasets show that the proposed ACANet is more competent for the brain tumor segmentation task than eight state-of-the-art segmentation methods.

To sum up, this work includes the following contributions.

- 1) A novel brain tumor segmentation network is proposed by considering ACA and prediction-aware decoding.

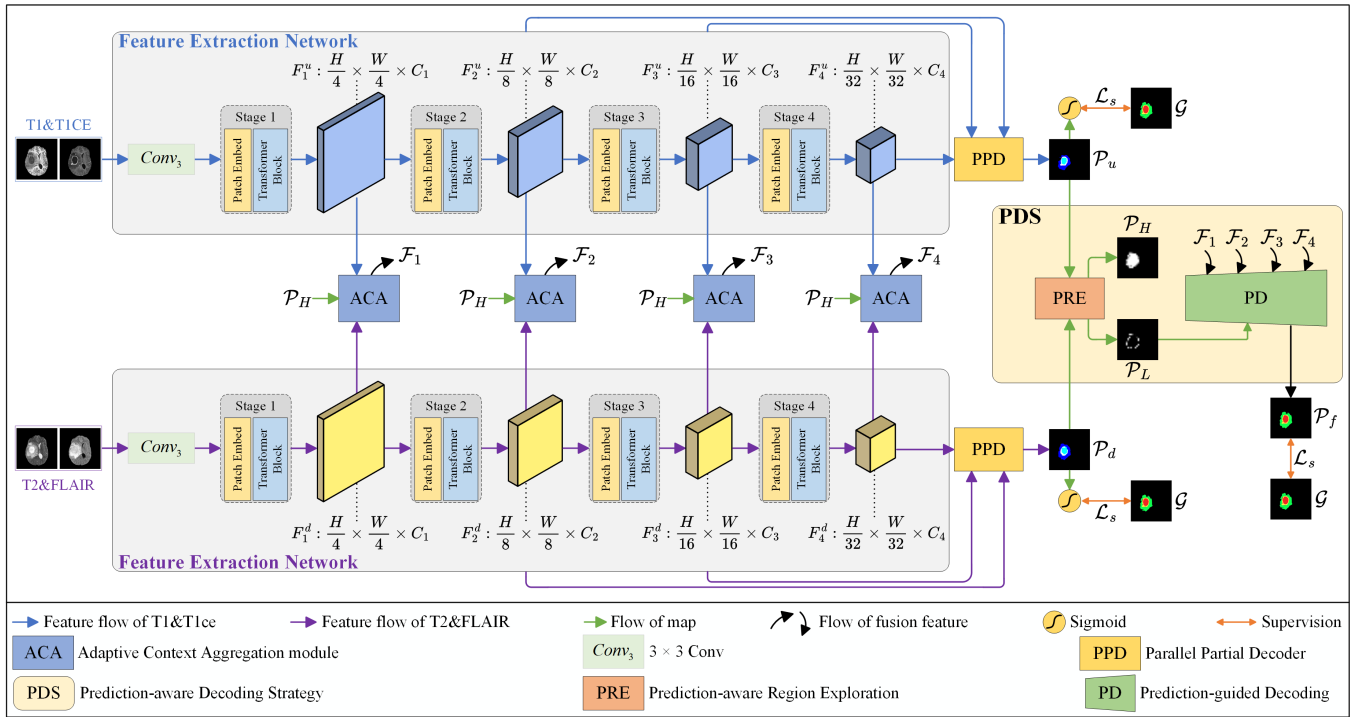


Fig. 1. Framework of the proposed ACANet that includes three key components: the feature extraction network, the ACA module, and the PDS. PDS contains a PRE module and a PD module. ACANet first employs the two parallel feature extraction networks to extract features for two modality pairs. Subsequently, the PPD module is utilized to aggregate features of top-three layers, resulting in two temporal predictions $\mathcal{P}_u$ and $\mathcal{P}_d$ for the upper and lower branches. These temporal predictions are used for generating $\mathcal{P}_H$ and $\mathcal{P}_L$ through the PRE module. The ACA module is then used to fuse the multimodal features in a multiscale manner with the guidance of $\mathcal{P}_H$. The fused features are further fed into the PD module through skip connection to decode in a stepwise manner with the guidance of $\mathcal{P}_L$, resulting the final prediction $\mathcal{P}_f$. For the sake of concise layout, we use short curved arrows to represent the skip connections. $\mathcal{G}$ denotes the ground truth.

The proposed network can efficiently cope with the challenges of large shape diversity and blurred boundary of tumors.

- 2) A novel ACA module is proposed to bridge the gaps between different modalities. Different from existing works that fuse multimodal features directly through the concatenation operation or adaptively through the weighting manner, our proposed ACA module incorporates the refined multimodal features in a multiscale way with the guidance of a united temporal prediction, enhancing the tumor-aware representation.
- 3) A novel PDS is introduced to cope with the complex boundary interactions in brain tumor segmentation. Different from existing methods that leverage an auxiliary task of edge detection, our PDS adaptively mines the boundary cues from the difference of two temporal predictions and uses these boundary cues to guide the decoding stage, yielding accurate segmentation results.

II. METHODOLOGY

A. Architecture Overview

As depicted in Fig. 1, our proposed ACANet follows an end-to-end framework and comprises three main components: the feature extraction network, the ACA module, and the PDS. It takes four image modalities of MRIs as the inputs and outputs three predictions, i.e., two temporal predictions ($\mathcal{P}_u$ and $\mathcal{P}_d$) and a final prediction $\mathcal{P}_f$. The four imaging

modalities are categorized into two pairs (i.e., T1 and T1CE, and T2 and FLAIR) according to their correlation, and these pairs serve as the inputs for the two parallel feature extraction networks, respectively. In this study, we take the popular PVTv2-B2 [52] (pretrained on ImageNet-1K [53]) as the feature extraction network because it creates hierarchical feature mappings and exhibits outstanding performance in various visual tasks.

To fully mine multimodal information, our ACANet includes two parallel branches, each of which corresponds to a special image pair (T1 and T1CE, or T2 and FLAIR). With the feature extraction network of each branch, we can obtain multiscale features $F_i^j \in \mathbb{R}^{C_i \times H/2^i \times W/2^i}$ ($i \in \{1, 2, 3, 4\}$, $j \in \{u, d\}$) by feeding the image pairs into the network. H and W denote the height and weight of the input, and C_i ($i \in \{1, 2, 3, 4\}$) is the channel dimension of features in the i th stage of the network. Since the input contains two modalities, F_i^j reflect the multimodal information. After extracting features F_i^u and F_i^d from the upper and lower feature extraction networks, we utilize an ACA module at each stage with the guidance of the united prediction $\mathcal{P}_H$ from the prediction-aware region exploration (PRE) module to fuse them adaptively, resulting in $\mathcal{F}_i$. $\mathcal{F}_i$ is further used as the input of the PDS through the skip connection. Specifically, PDS consists of a PRE module and a prediction-guided decoding (PD) module. The former aims to explore united prediction regions $\mathcal{P}_H$ and low-confidence prediction regions $\mathcal{P}_L$ via analysis of the relation between two temporal predictions

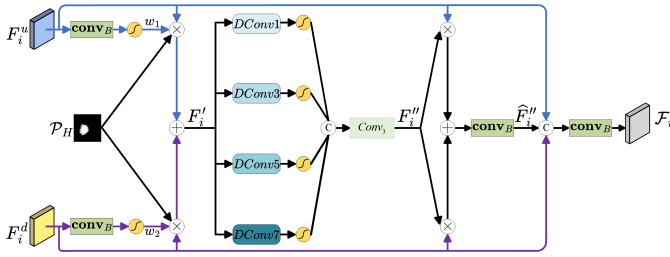


Fig. 2. Framework of the proposed ACA module. Conv_3 means the 3×3 convolution operation, and conv_B denotes the convolution block.

$\mathcal{P}_u$ and $\mathcal{P}_d$ from the upper and lower branches. Considering that high-stage features contain rich semantic information, we aggregate the features of the top-three layers using the parallel partial decoder (PPD) module [54] to generate $\mathcal{P}_u$ and $\mathcal{P}_d$ in the upper and lower branches, respectively. Here, only the temporal predictions of the whole tumor (WT) from both branches are used for generating $\mathcal{P}_H$ and $\mathcal{P}_L$, because we hope that the network can find all candidate tumors. The PD module decodes the skip-connected features $\mathcal{F}_4 \sim \mathcal{F}_1$ in a stepwise manner with the guidance of $\mathcal{P}_L$ and outputs the final prediction $\mathcal{P}_f$. In what follows, we will detail the proposed ACA module and PDS, followed by the introduction of the used loss function.

B. ACA Module

Despite recording rich and relevant information of the same scene, multimodal data exhibit differences due to variations in imaging principles, resulting in distinct data representations. Such differences often confuse the model when mining discriminative representations for accurate segmentation. Therefore, how to effectively extract discriminative representations is a challenging task. Given that directly concatenating multimodal features may mask the discriminative information of each modality, we propose an ACA module to extract and fuse complementary information from multimodal features.

Fig. 2 illustrates the structure of the proposed ACA module. It takes the side outputs (F_i^u and F_i^d) of two feature extraction networks and the united prediction $\mathcal{P}_H$ of the PRE module as the inputs and outputs the fused feature $\mathcal{F}_i$. To be specific, we first process F_i^u and F_i^d with a 3×3 convolution block (that consists of a 3×3 convolution and a batch normalization), followed by a Sigmoid function, to obtain the weighting maps $w_1, w_2 \in [0, 1]$. Here, a dual-branch structure is utilized without any parameter sharing. Then, we multiply the weighting maps with the input to explore discriminative information, respectively. Next, we linearly combine the two weighted input features and multiply the resultant feature map with the united prediction $\mathcal{P}_H$ provided by the PDS (introduced later in Section II-C) to help the network focus more on the tumor regions

$$\begin{cases} w_1 = \sigma(\text{conv}_B(F_i^u)) \\ w_2 = \sigma(\text{conv}_B(F_i^d)) \end{cases} \quad (1)$$

$$F_i' = (w_1 \cdot F_i^u + w_2 \cdot F_i^d) \cdot \mathcal{P}_H \quad (2)$$

where $\text{conv}_B(\cdot)$ denotes the convolution block, and $\sigma(\cdot)$ is the Sigmoid function. Before the multiplication operation,

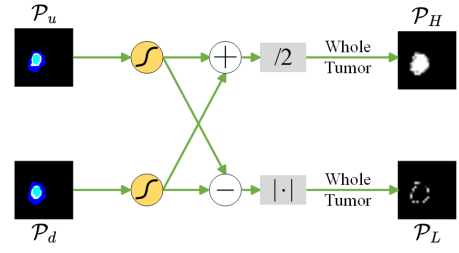


Fig. 3. Framework of the proposed PRE module.

$\mathcal{P}_H$ is resized to the same channel size as F_i^u and F_i^d . After that, F_i' is processed by a group of parallel dilated convolutions with the dilated rates of 1, 3, 5, and 7 to represent the tumor-aware features into a multiscale manner. A Sigmoid function is used after each dilated convolution. These multiscale features are concatenated along the channel direction and further processed by a 3×3 convolution. Next, the resultant feature map F_i'' is multiplied with the two inputs, respectively, and the multiplication results are added together and further processed by a 3×3 convolution block, resulting in $\hat{F}_i''$. Finally, we concatenate $\hat{F}_i''$, F_i^u , and F_i^d and feed the concatenated feature into another 3×3 convolution block to generate the output $\mathcal{F}_i$ of the ACA module. The above operations can be formulated as

$$\hat{F}_i'' = \text{conv}_B(F_i'' \cdot (F_i^u + F_i^d)) \quad (3)$$

$$\mathcal{F}_i = \text{conv}_B(\hat{F}_i'' \odot F_i^u \odot F_i^d) \quad (4)$$

where $\odot$ means the concatenation operation between two features along the channel direction.

C. Prediction-Aware Decoding Strategy

After obtaining tumor-aware features from the ACA modules in the encoder, our next step is to decode them to generate accurate segmentation mask. For this purpose, we propose a PDS, which comprises a PRE module and a prediction-guided decoding module.

1) *Prediction-Aware Region Exploration*: Since different image modalities represent the tumor information from different aspects, the two parallel branches in Fig. 1 may have different segmentation predictions because of the different fed image pairs. Generally, the central regions of the tumor are more easily detected by the segmentation model, while the edge regions are more uncertainly. In view of this, the proposed PRE module aims to generate both the united prediction regions $\mathcal{P}_H$ and low-confidence prediction regions $\mathcal{P}_L$. Fig. 3 illustrates the structure of the proposed PRE module. Specifically, for the two temporal WT predictions $\mathcal{P}_u$ and $\mathcal{P}_d$ from two branches, we process them with a Sigmoid function and generate $\mathcal{P}_H$ and $\mathcal{P}_L$ through addition and subtraction operations

$$\mathcal{P}_H = (\sigma(\mathcal{P}_u) + \sigma(\mathcal{P}_d))/2 \quad (5)$$

$$\mathcal{P}_L = |\sigma(\mathcal{P}_u) - \sigma(\mathcal{P}_d)|. \quad (6)$$

The reason why we select the predictions of WT to explore the prediction-aware regions is that WT has the larger area than its subregions, e.g., tumor core (TC) and enhanced tumor (ET), helping the network focus on the whole

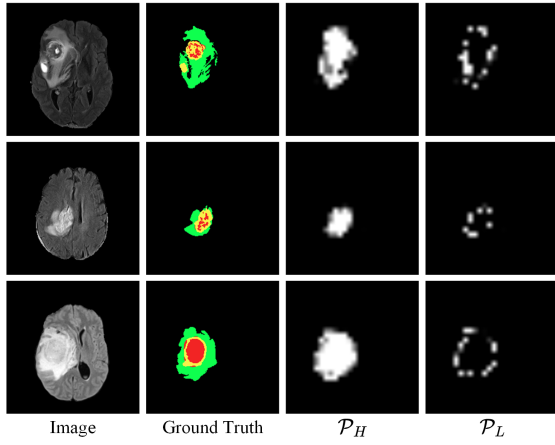


Fig. 4. Visualization of $\mathcal{P}_H$ and $\mathcal{P}_L$. As can be seen, $\mathcal{P}_H$ presents the candidate tumor regions, while $\mathcal{P}_L$ provides the boundary cues.

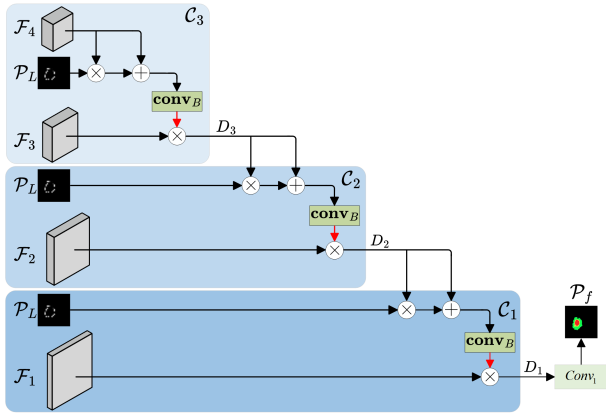


Fig. 5. Framework of the proposed PD module. Conv_1 means the 1×1 convolution and conv_B denotes the convolution block, which consists of a 3×3 convolution and a batch normalization.

lesion areas. According to (5) and (6), $\mathcal{P}_H$ includes all possible lesion regions, while $\mathcal{P}_L$ contains prediction uncertainty, which usually happens on the boundary regions. For convenient understanding of the effect of $\mathcal{P}_H$ and $\mathcal{P}_L$, we present some visualization maps in Fig. 4.

2) *Prediction-Guided Decoding*: Segmentation model usually fails to accurately recognize the tumor boundary due to the low contrast between foreground and background. To help the network concentrate more on boundary regions, we propose a PD module that decodes the features $\mathcal{F}_1 \sim \mathcal{F}_4$ from the ACA modules in a stepwise manner through the assistance of $\mathcal{P}_L$. As illustrated in Fig. 5, the proposed PD module contains three connected structurally identical components $\mathcal{C}_k$ ($k \in \{3, 2, 1\}$), named in reverse order for convenient expression, and each component takes $\mathcal{F}_i$, $\mathcal{P}_L$, and the output D_{i+1} of the adjacent high component as the inputs and generates an enhanced feature D_i . Taking the first component $\mathcal{C}_3$ (which is closest to the top stage $\mathcal{F}_4$) as an example, the feature $\mathcal{F}_4$ is first multiplied with $\mathcal{P}_L$ and then added with the multiplied results. Here, we utilize $\mathcal{F}_4$ as the output of the adjacent high component. Next, the resultant feature is processed by a convolution block and multiplied with $\mathcal{F}_3$, resulting in the enhanced feature D_3 . The above process can be formulated by

$$D_i = \text{conv}_B(D_{i+1} \cdot \mathcal{P}_L + D_{i+1}) \cdot \mathcal{F}_i, \quad i \in \{1, 2, 3\} \quad (7)$$

where $\mathcal{F}_4$ is used to play the role of D_4 . In (5), $\mathcal{P}_L$ is resized to the same spatial size of D_{i+1} using the downsampling operation, and $\text{conv}_B(D_{i+1} \cdot \mathcal{P}_L + D_{i+1})$ is resized to the same size of $\mathcal{F}_i$ using the upsampling operation. The output D_3 of the first component is further served as the input of the second component, which outputs D_2 . By connecting these three components (i.e., $\mathcal{C}_3$, $\mathcal{C}_2$, and $\mathcal{C}_1$) in a stepwise manner, we can finally obtain the enhanced feature D_1 . The final prediction $\mathcal{P}_f$ is generated by processing D_1 with a 1×1 convolution operation. The above procedure can be formalized as

$$\mathcal{P}_f = \sigma(\text{Con}_1(\mathcal{C}(\mathcal{C}(\mathcal{C}(\mathcal{F}_4, \mathcal{P}_L, \mathcal{F}_3), \mathcal{P}_L, \mathcal{F}_2), \mathcal{P}_L, \mathcal{F}_1))) \quad (8)$$

where $\text{Con}_1(\cdot)$ is the 1×1 convolution operation, and $\mathcal{C}(\cdot, \cdot, \cdot)$ denotes the operations described in (5).

D. Loss Function

The proposed ACANet includes three supervision signals corresponding to the two temporal predictions and a final prediction, as shown in Fig. 1. We assign each prediction a hybrid loss function and formulate the whole loss function of our ACANet as

$$\mathcal{L} = \mathcal{L}_s(\mathcal{P}_u, \mathcal{G}) + \mathcal{L}_s(\mathcal{P}_d, \mathcal{G}) + \mathcal{L}_s(\mathcal{P}_f, \mathcal{G}) \quad (9)$$

where $\mathcal{G}$ denote the ground-truth masks that include the region information of WT, TC, and ET. $\mathcal{L}_s$ is the basic loss function for the segmentation task

$$\mathcal{L}_s = \mathcal{L}_D + \mathcal{L}_C \quad (10)$$

where $\mathcal{L}_D$ and $\mathcal{L}_C$ are the classical Dice loss and cross-entropy loss. During loss calculation, we resize the three predictions to the same size of the ground-truth mask and process them with a Sigmoid function.

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Setup

1) *Dataset*: In this study, the BraTS 2020 [4] and BraTS 2021 [55] datasets are selected to evaluate and compare the effectiveness of segmentation methods. These datasets encompass a wide variety of brain tumor cases, including different tumor types, sizes, and locations. Specifically, BraTS 2020 and BraTS 2021 contain 369 and 1251 multi-modal MR scans, respectively, and each scan includes four different modalities, i.e., T1, T1CE, T2, and FLAIR. All scans are meticulously segmented by experienced annotators and neuro-radiologists, providing annotations that encompass the WT, ET, and TC. Some examples are shown in Fig. 6. Following previous works [3], [47], we focus on 2-D lesion segmentation and extract 2-D slices from the axial-plane of the original data. We crop each 2-D slice to 224×224 pixels to remove unnecessary background information, and normalize it within the range of $[0, 1]$. Due to the unavailability of the test part in BraTS 2020 and BraTS 2021, we conduct experiments solely using the training part, which is divided into training and testing sets based on the original subject ID of the slices, with an allocation of 80% for training and 20% for testing, respectively.

2) *Evaluation Metrics*: We use four widely recognized evaluation metrics in the field of medical image segmentation

TABLE I
QUANTITATIVE COMPARISONS OF MAINSTREAM SEGMENTATION METHODS ON BraTS 2020

Methods	FLOPs (G)	Dice $\uparrow$				IoU $\uparrow$				Sensitivity $\uparrow$				HD95 $\downarrow$			
		WT	TC	ET	Mean	WT	TC	ET	Mean	WT	TC	ET	Mean	WT	TC	ET	Mean
U-Net [20]	41.94	0.8902	0.8207	0.8131	0.8413	0.7965	0.6906	0.6723	0.7198	0.8908	0.8421	0.8866	0.8732	2.9762	2.4300	2.1048	2.5037
U-Net++ [22]	106.20	0.8964	0.8204	0.8392	0.8520	0.8075	0.6986	0.7147	0.7403	0.8717	0.7458	0.7785	0.7987	2.7990	1.6236	2.0128	2.1451
Attention U-Net [21]	34.88	0.8950	0.8169	0.8466	0.8528	0.8061	0.6927	0.7249	0.7412	0.8617	0.7710	0.8329	0.8217	2.7806	1.7990	2.2034	2.2610
U-Net3+ [23]	153.25	0.9034	0.8320	0.8555	0.8636	0.8238	0.7124	0.7474	0.7612	0.8818	0.7869	0.8340	0.8342	2.7700	1.9882	1.6263	2.1281
CENet [56]	6.85	0.9067	0.8487	0.8366	0.8640	0.8293	0.7371	0.7191	0.7618	0.8965	0.8143	0.8248	0.8452	2.7779	1.9564	1.6483	2.1275
Swin-U-Net [27]	5.93	0.9044	0.8402	0.8558	0.8668	0.8254	0.7245	0.7479	0.7659	0.8750	0.7947	0.8170	0.8289	2.7612	1.9224	1.5335	2.0724
CaraNet [57]	8.81	0.8869	0.8274	0.7733	0.8292	0.7968	0.7056	0.6304	0.7109	0.8872	0.8207	0.8259	0.8446	2.9899	1.9704	1.7627	2.2410
IVD-Net [58]	76.05	0.9062	0.8382	0.8567	0.8670	0.8241	0.7242	0.7409	0.7631	0.8758	0.7876	0.8211	0.8282	2.7318	1.6225	1.9985	2.1176
ACANet (Ours)	23.46	0.9117	0.8647	0.8509	0.8758	0.8376	0.7617	0.7405	0.7800	0.9134	0.8556	0.8721	0.8804	2.7452	2.0193	1.7336	2.1660

TABLE II
QUANTITATIVE COMPARISONS OF MAINSTREAM SEGMENTATION METHODS ON BraTS 2021

Methods	Dice $\uparrow$				IoU $\uparrow$				Sensitivity $\uparrow$				HD95 $\downarrow$			
	WT	TC	ET	Mean	WT	TC	ET	Mean	WT	TC	ET	Mean	WT	TC	ET	Mean
U-Net [22]	0.9183	0.8622	0.8868	0.8891	0.8489	0.7579	0.7967	0.8012	0.9018	0.9217	0.8833	0.9023	2.6560	2.2674	1.7581	2.2272
U-Net++ [22]	0.9088	0.8956	0.8801	0.8948	0.8328	0.8109	0.7859	0.8099	0.8649	0.8856	0.9152	0.8886	2.6661	1.9180	1.8533	2.1458
Attention U-Net [21]	0.9160	0.8821	0.8865	0.8949	0.8450	0.7891	0.7961	0.8100	0.8812	0.9218	0.9028	0.9019	2.6310	2.1737	1.8947	2.2331
U-Net3+ [23]	0.9118	0.9060	0.8849	0.9009	0.8379	0.8281	0.7935	0.8199	0.8715	0.8704	0.8677	0.8699	2.6594	1.7091	1.6827	2.0171
CENet [56]	0.9173	0.9095	0.8693	0.8987	0.8473	0.8339	0.7688	0.8167	0.9047	0.8779	0.8744	0.8857	2.6800	1.5584	1.5907	1.9430
Swin-U-Net [27]	0.9234	0.9016	0.8976	0.9075	0.8576	0.8208	0.8143	0.8309	0.8912	0.8545	0.8755	0.8738	2.5712	1.5728	1.5049	1.8830
CaraNet [57]	0.9062	0.9021	0.8034	0.8705	0.8285	0.8216	0.6714	0.7738	0.9104	0.8887	0.9176	0.9056	2.8261	1.6093	1.8161	2.0838
IVD-Net [58]	0.9252	0.8882	0.8881	0.8980	0.8609	0.7988	0.7865	0.8154	0.9182	0.8992	0.9126	0.9100	2.5824	2.1509	2.0508	2.2614
ACANet (Ours)	0.9318	0.9225	0.8862	0.9135	0.8723	0.8561	0.7957	0.8413	0.9129	0.9031	0.8859	0.9006	2.5102	1.6469	1.6866	1.9479

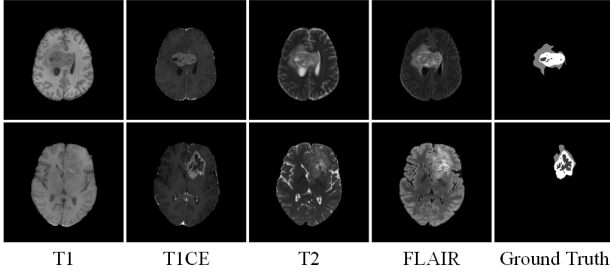


Fig. 6. Examples from the BraTS 2021 dataset. From left to right, the images are captured in the imaging modality of T1, T1CE, T2, and FLAIR, respectively. The segmentation ground-truth is presented at the last column.

to present a comprehensive quantitative analysis of segmentation methods. These metrics include Dice similarity coefficient (Dice), Intersection over Union (IoU), Sensitivity, and 95% Hausdorff distance (HD95). Generally, a superior segmentation method produces higher values of Dice, IoU, and Sensitivity, while lower value of HD95. The four evaluation indicators are defined as follows:

$$\text{Dice} = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FP} + \text{FN}} \quad (11)$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \quad (12)$$

$$\text{HD95} = \max \left\{ \max_{a \in A} \min_{b \in B} d(a, b), \max_{b \in B} \min_{a \in A} d(b, a) \right\} \quad (13)$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (14)$$

where TP, FP, and FN denote the counts of true positive, false positive, and false negative pixels, respectively. A and B are the two sets of tumor boundary pixels corresponding to predicted results and ground truth annotations, respectively. $d(\cdot, \cdot)$ denotes the Euclidean distance between two pixels.

3) *Implementation Details:* In this study, we train our ACANet on an Ubuntu server equipped with one NVIDIA 3090 GPU. Our network's parameters are fine-tuned

with the SGD optimizer. During the training phase, we set the initial learning rate to $1e-2$ and implement the “Poly” strategy for gradual reduction. The batch size is configured as 12, and we train our network for a total of 100 epochs.

B. Comparisons With State-of-the-Arts

To illustrate the effectiveness of our ACANet, we select eight state-of-the-art segmentation methods in our comparison experiments, including UNet [20], UNet++ [22], Attention U-Net [21], U-Net3+ [23], CENet [56], Swin-U-Net [27], Caranet [57], and IVD-Net [58]. Among them, IVD-Net belongs to the multimodal methods, and the rest belong to the single-modal methods. Since the single-modal methods requires only one input, we cascade the four imaging modalities and then feed them into the segmentation network through a 3×3 convolution operation. In addition, the number of output channels is extended to three to correspond to the predictions of WT, TC, and ET.

1) *Quantitative Analysis:* We present the quantitative results of different segmentation methods on the BraTS 2020 and BraTS 2021 datasets in Tables I and II. For convenient observation, we mark the best result of each evaluation metric on the three segmentation tasks in bold. Following previous works, the mean value of three tasks is also reported. As seen, our proposed method achieves impressive results and shows obvious superiority over the eight competing methods on most of the evaluation metrics. Specifically, when tested on the BraTS 2020 dataset, the proposed ACANet obtains better results in Dice, IoU, and Sensitivity for WT and TC. For example, the obtained Dice values for WT and TC are 0.9117 and 0.8647, respectively, bringing approximately 0.50% and 1.60% gains than those of the second best method (CENet). On BraTS 2021, ACANet improves its Dice values to 0.9318 and 0.9225 and achieves the IoU values of 0.8723 and 0.8561 when segmenting WT and TC, respectively. At the WT level, the Dice and the IoU values are higher than those of the second-ranked IVD-Net by 0.66% and 1.14%, respectively.

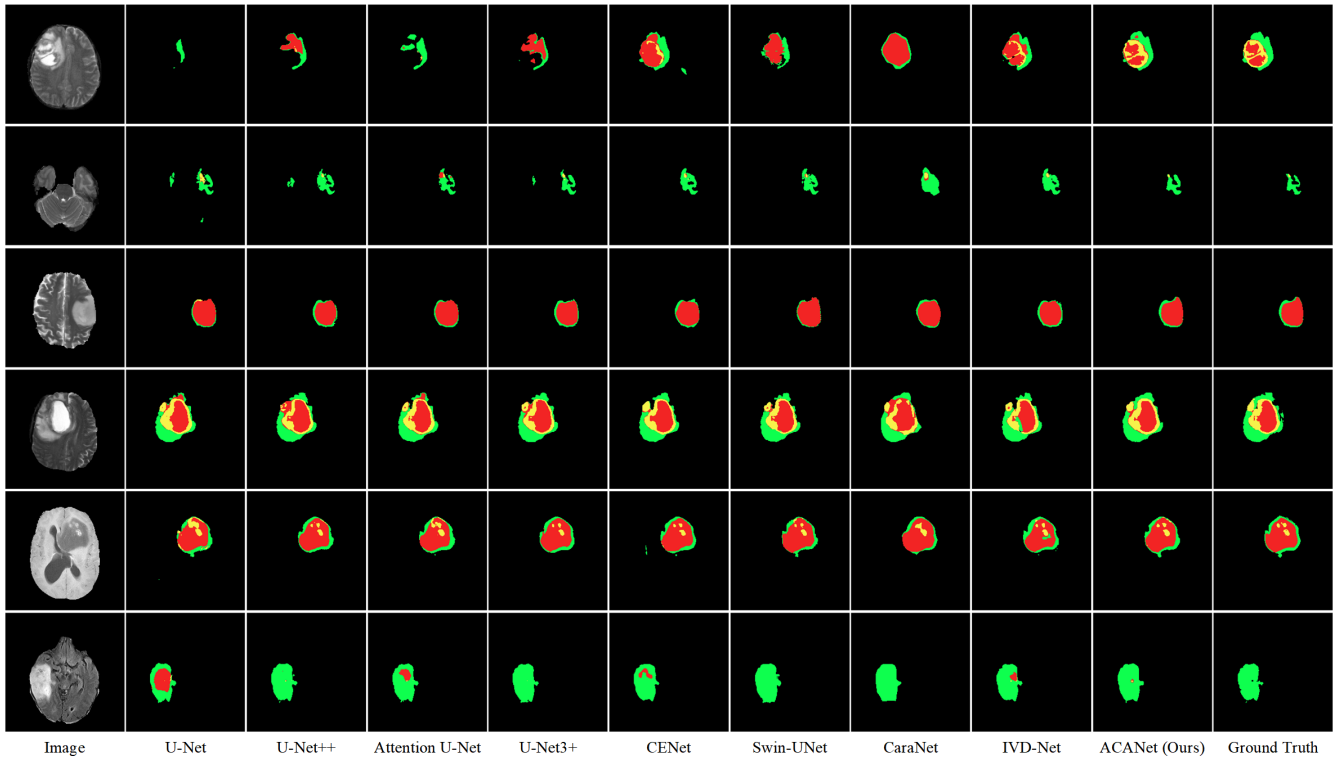


Fig. 7. Visual comparisons between our ACANet and other eight competing methods on the BraTS 2020 dataset (green, yellow, and red regions represent the WT, TC, and ET, respectively).

However, although ACANet ranks second and first on BraTS 2020 and BraTS 2021 in terms of the HD95 metric, respectively, when segmenting WT, it does not achieve very competitive performance in terms of the HD95 metric when segmenting TC and ET. This may be because our ACANet mainly focuses on exploring the prediction-aware region at the WT level with the assistance of $\mathcal{P}_H$ and $\mathcal{P}_L$, while ignoring the PRE at the TC and ET levels. Despite this, our ACANet ranks first 11 times and second 3 times in a total of 24 comparisons, showing higher performance stability than competing methods. A possible reason for this is that our ACANet adaptively fuses multimodal data using a proposed ACA module with the guidance of the united temporal prediction $\mathcal{P}_H$, which mainly includes the whole candidate tumor regions yet sometimes ignores the specific characteristics of small lesion, i.e., ET. In addition, the proposed PD strategy contributes to detecting the outline of the candidate tumor regions. Benefiting from these two mechanisms, our ACANet can detect the large lesions (WT and TC) more accurately than the small lesion (ET).

We also compare these methods in terms of floating point operations per seconds (FLOPs). As shown in Table I, compared with the competing methods, ACANet maintains computational efficiency in the form of FLOPs within a reasonable range. Despite this, ACANet's leading position on evaluation indicators indicates its superior segmentation precision. It is worth noting that our current study mainly focuses on proposing a high-performance segmentation model as it is very important to the disease diagnosis and subsequent surgical planning. We leave the reduction of computational complexity as a future work.

2) *Qualitative Analysis*: To provide a more intuitive comparison of segmentation performance, we present the visualization results of competing methods and our proposed ACANet on the BraTS datasets in Figs. 7 and 8. Based on these visualization results, it is clear that single-modal methods usually fail in accurately segmenting challenging brain tumors (e.g., complex interactions between subregions in the first and last four rows of Fig. 7 and the first row of Fig. 8, complex shape and size in the second and fourth rows of Fig. 7 and the first three rows and last row of Fig. 8, and blurred boundaries in the fourth row of Fig. 7 and the third row of Fig. 8), resulting in either underestimated regions, overestimated regions, or the hybrid. A possible reason for this is that single-modal methods do not consider the distinctive information inherent in brain tumor segmentation between different modalities. In contrast, our ACANet achieves more promising segmentation results than other methods, exceling in both WT and TC segmentation and being competitive in ET segmentation. Specifically, compared with the multimodal method (IVD-Net), our ACANet can achieve more accurate results. It generates sharper boundaries and produces more accurate segmentation results for lesions with various shapes and sizes. A possible reason for this is that we take $\mathcal{P}_H$ and $\mathcal{P}_L$ generated by the PRE module as the guidance information to help ACANet focus on the lesion and its boundary regions.

C. Ablation Study

This section conducts several ablation experiments to investigate the contributions of some design concepts in ACANet, including the usage of ACA module and guidance

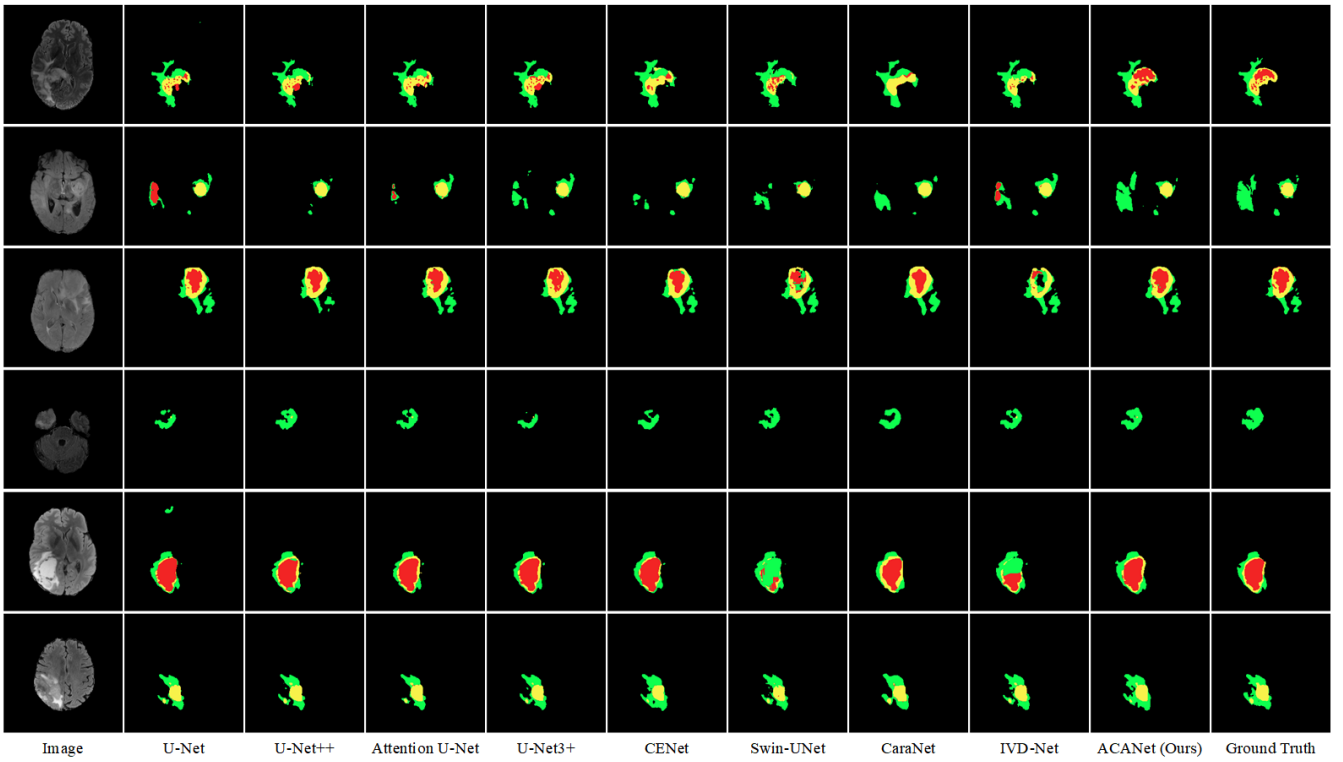


Fig. 8. Visual comparisons between our ACANet and other eight competing methods on the BraTS 2021 dataset (green, yellow, and red regions represent the WT, TC, and ET, respectively).

TABLE III
QUANTITATIVE RESULTS OF ABLATION STUDIES

Datasets	BraTS 2020				BraTS 2021			
	Dice $\uparrow$	IoU $\uparrow$	Sensitivity $\uparrow$	HD95 $\downarrow$	Dice $\uparrow$	IoU $\uparrow$	Sensitivity $\uparrow$	HD95 $\downarrow$
w/o ACA	0.8585	0.7533	0.8544	2.2921	0.9084	0.8329	0.8992	2.0162
w/o $\mathcal{P}_H$	0.8690	0.7696	0.8593	2.2127	0.9106	0.8364	0.9106	1.9837
w/o $\mathcal{P}_L$	0.8708	0.7724	0.8535	2.1291	0.9095	0.8346	0.8983	2.0422
ACANet	0.8758	0.7800	0.8804	2.1660	0.9135	0.8413	0.9006	1.9479

information ($\mathcal{P}_H$ and $\mathcal{P}_L$). In each experiment, we only remove the considered module or the guidance information, while keeping the remaining parts of the network unchanged to learn a new model. The experimental setup is the same as in Section III-A. Due to space limitation, we only report the mean results on three segmentation tasks, i.e., WT, TC, and ET segmentation.

1) *Effectiveness of ACA*: The proposed ACA module aims to adaptively fuse multimodal features extracted by two parallel feature extraction network. To achieve this, the weighting mechanism, multiscale feature extraction, and guidance from the united prediction region $\mathcal{P}_H$ are used. Here, we remove the ACA module and fuse the multimodal features directly via the concatenation operation. It is clear that, the removal of our proposed ACA module causes an obvious performance drop, as shown in Table III. For example, it has brought approximately 1.73% and 0.51% decrease of Dice, 2.67% and 0.84% decrease of IoU, 2.60% and 0.14% decrease of Sensitivity, and 0.1261 and 0.0683 increase of HD95 in BraTS 2020 and BraTS 2021, respectively. This demonstrates that the proposed ACA module contributes to discriminative multimodal feature fusion for accurate brain tumor segmentation.

2) *Effectiveness of $\mathcal{P}_H$* : The united prediction region $\mathcal{P}_H$ generated by the PRE module is used to guide the fusion of multimodal features in the ACA module, which can assist the network concentrate on possible tumor regions. Here, we remove $\mathcal{P}_H$ from the ACA module and keep the remaining parts of ACA module unchanged. From the comparisons between the second and last rows of Table III, it is clear that $\mathcal{P}_H$ plays a positive role in segmentation performance improvement. For example, with the guidance of $\mathcal{P}_H$, the Dice value increases by approximately 0.68% and 0.29% in BraTS 2020 and BraTS 2021, respectively.

3) *Effectiveness of $\mathcal{P}_L$* : In our ACANet, $\mathcal{P}_L$ aims to provide the boundary cues (usually considered to be error-prone regions). Taking these boundary cues as guidance, the proposed PD module can decode the skip-connected features from ACA modules in a stepwise manner, generating accurate segmentation with clear boundaries. Here, we remove $\mathcal{P}_L$ to see how the performance of the network changes. In this case, the network's ability to perceive tumor boundaries decreases. Without any surprises, the result will decrease in the absence of guidance from $\mathcal{P}_L$. As shown in the last two rows of Table III, the Dice value decreases by 0.50% and 0.40% and the IoU value decreases by 0.76% and 0.67% on the BraTS 2020 and BraTS 2021 datasets, respectively.

4) *Visual Comparisons*: We present some visual results from our ablation study in Fig. 9. Compared to the model without the ACA module or without the guidance from $\mathcal{P}_H$ or $\mathcal{P}_L$, the standard ACANet provides more impressive predictions across the WT, TC, and ET categories. For instance, for WT, our predictions more closely align with the

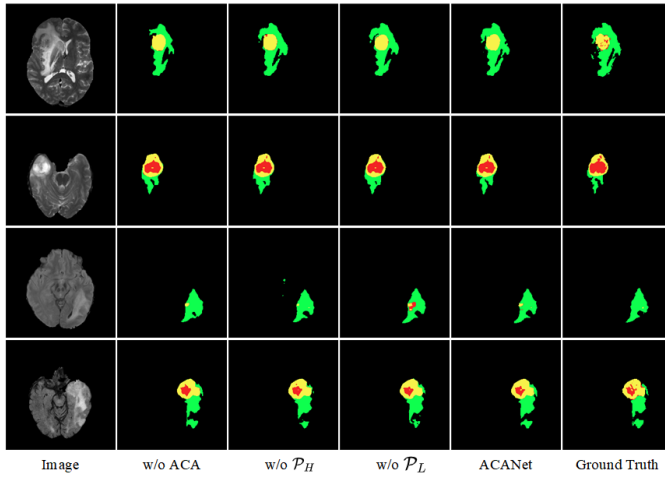


Fig. 9. Visual comparisons between our proposed ACANet and its variants, where the first two rows are from BraTS 2020, and the last two rows are from BraTS 2021.

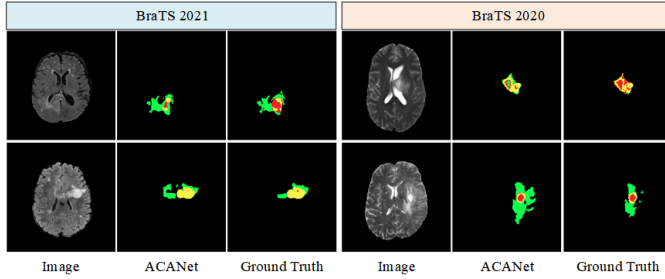


Fig. 10. Some failure cases of our method.

ground truth, with particular improvements observed in boundaries details and tumor completeness. Moreover, compared to its variants, ACANet achieves more accurate prediction for tumors with complex boundaries, as illustrated in the first row of Fig. 9.

D. Discussion

Based on both quantitative and qualitative analyses on two datasets, it is evident that our proposed ACANet outperforms eight competing methods in dealing with the WT. However, ACANet's performance in segmenting the TC and ET is not as notable, especially with small ETs. As can be seen from the first row of Fig. 10, our ACANet performs better in WT segmentation but does not perform well in coping with complex TC and ET. This phenomenon may stem from the fact that $\mathcal{P}_H$ and $\mathcal{P}_L$ in ACANet are primarily designed at the WT level, with the intention of encompassing all potential tumor regions. More specifically, $\mathcal{P}_H$ and $\mathcal{P}_L$, containing WT cues, improve the network's focus on the WT level but possibly detracting from the attention to TC and ET levels during the ACA module and decoding stages. Since $\mathcal{P}_H$ is used to capture all potential tumor regions, our proposed network may misidentify complex tumors that have high similarity to its neighboring tissues. Some failure cases are shown in the last row of Fig. 10.

To move this field more forward, some efforts will be made in the future. On the one hand, our future research will explore

how to improve the segmentation results at the TC and ET levels on the premise of accurately segmenting WT. For this purpose, advanced semantic exploration strategy is considered to enhance the network's ability to cope with tumors with different sizes and shapes. On the other hand, reducing the computational complexity is also our future direction. One possible solution is utilizing a single-branch feature extraction approach for processing different modalities and developing specialized modules for retaining the unique features of each modality. Such a design concept will not only improve the processing efficiency but also ensure that the unique features inherent to each modality are retained and effectively utilized.

IV. CONCLUSION

In this article, we address the challenging multimodal brain tumor segmentation task by proposing a novel ACANet. The proposed ACANet includes two key components, namely ACA module and PDS. The ACA module aims to fuse multimodal features adaptively to bridge the gaps between different modalities, thereby improving the tumor-aware feature representation capability. The PDS adaptively explores the boundary cues from temporal predictions and utilizes them to guide the decoding stage for accurate segmentation. Through the natural cooperation of the proposed ACA module and PDS, our ACANet can segment the brain tumor effectively. Experimental results on two datasets show that ACANet outperforms eight state-of-the-art methods.

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Guanghui Yue (Member, IEEE) received the B.S. degree in communication engineering and the Ph.D. degree in information and communication engineering from Tianjin University, Tianjin, China, in 2014 and 2019, respectively.

He was a joint Ph.D. Student with the School of Computer Science and Engineering, Nanyang Technological University, Singapore, from September 2017 to January 2019. He is currently an Associate Professor with the School of Biomedical Engineering, Health Science Center, Shenzhen University, Shenzhen, China. His research interests include medical image analysis, bioelectrical signal processing, image quality assessment, 3-D image visual discomfort prediction, pattern recognition, and machine learning.



Shangjie Wu received the B.S. degree in biomedical engineering from Shenzhen University, Shenzhen, China, in 2023, where he is currently pursuing the master's degree in biomedical engineering.

His research interests include deep learning and medical image analysis.



Jie Du (Member, IEEE) received the B.S. degree in computer science and technology from the University of Shihezi, Xinjiang, China, in 2013, and the Ph.D. degree from the Department of Computer and Information Science, University of Macau, Macau, China, in 2019.

She is currently an Assistant Professor with the School of Biomedical Engineering, Shenzhen University Medical School, Shenzhen University, Shenzhen, China. Her research interests include machine learning methods and medical image processing.



Tianwei Zhou (Associate Member, IEEE) received the B.S. degree in automation and the Ph.D. degree in control science and engineering from Tianjin University, Tianjin, China, in 2014 and 2019, respectively.

She was a joint Ph.D. Student with the Department of Electrical and Computer Engineering, National University of Singapore, Singapore, from August 2017 to August 2018. She is currently an Assistant Professor with the College of Management, Shenzhen University, Shenzhen. Her current research interests include event-triggered control, intelligent scheduling, image processing, and medical image analysis.



Bin Jiang (Member, IEEE) received the B.S. and M.S. degrees from Tianjin University, Tianjin, China, in 2013 and 2016, respectively, and the Ph.D. degree in information and communication engineering from Tianjin University in 2020.

He was a joint Ph.D. Student at the Security and Optimization for Networked Globe Laboratory, Department of Electrical Engineering and Computer Science, Embry-Riddle Aeronautical University, Daytona Beach, FL, USA, from November 2017 to February 2018 and from October 2018 to October 2019. He was a Post-Doctoral Researcher with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China, from August 2020 to July 2022. He is currently an Associate Professor with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao, China. His major research interests include the Internet of Things, communications, and information security.



Tianfu Wang received the Ph.D. degree in biomedical engineering from Sichuan University, Chengdu, China, in 1997.

He is currently a Professor with Shenzhen University, Shenzhen, China. His current research interests include ultrasound image analysis, medical image processing, pattern recognition, and medical imaging.